VIJAYSINGH PUWAR

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SUMMARY

Cybersecurity engineer with systems and network operations experience at R.S. Infotech and the MTA. At the MTA, handles access requests and incidents on a 24/7 rail network. Outside work, builds and runs two production web platforms and fixes their security issues. CompTIA Security+, CompTIA CySA+, and Cisco CCNA certified.

EXPERIENCE

Emerging Talent Intern | Metropolitan Transportation Authority (MTA) New York, NY | Jun 2026 – Present

- Review Active Directory, DUO MFA, and Zscaler ZPA access requests with the security team against policy.
- Handle incidents in ServiceNow and escalate to senior staff or vendors such as Dell, HP, and Solari.
- Inventoried every device on the railway network and traced each one to its switch port.
- Troubleshoot Cisco Catalyst switches and OSPF and BGP routing at sites across the railway.
- Built the SIR IT request intake system in Power Automate and SharePoint and wrote its runbook.
- Wrote and deployed a Python service that corrects clock drift over NTP on a Rail Control Center workstation.

System Engineer | R.S. Infotech Vadodara, India | Feb 2023 – Aug 2024

- Ran 150+ Windows and Linux systems on premises and on AWS and Azure, including patching and scheduled changes.
- Managed Active Directory and Group Policy: MFA enrollment, privileged group reviews, onboarding, and offboarding.
- Investigated incidents in Splunk and tuned IDS/IPS and firewall rules to cut false positives.
- Triaged SonarQube, OWASP ZAP, Burp Suite, and Nessus findings and added security checks to GitLab CI.
- Wrote Python, PowerShell, and Bash scripts for compliance checks, log review, and inventory.
- Tested backup and disaster recovery, and prepared evidence for audits and client security questionnaires.

PROJECTS

Web Audio TV | Audiobook streaming platform, about 1,000 users webaudiotv.com

TypeScript, React, PostgreSQL, Supabase, Cloudflare Workers

- Found and fixed three production security flaws: an email queue the public API key could call, a storage bucket anyone could download from, and a leaderboard farming exploit.
- Found the farming account with SQL over raw listening events (credited 16 to 21 hours a day) and added daily caps.
- Moved audio behind a Cloudflare Worker that serves a file only with a valid signed ticket.
- Verify every database role's permissions against the real migrations with a 48-check access matrix.

Recap Verse | AI video SaaS with paying subscribers recapverse.com

Python, FastAPI, Next.js, MongoDB, Redis, Google Cloud Run

- Built and run it alone: 398 deployments and 2,019 automated tests.
- Encrypt stored user API keys with AES-GCM and check resource ownership on every route.
- Remediated a leaked API key and added gitleaks secret scanning to CI.

AWS Cloud Security Lab github.com/VijaysinghPuwar/Configuring-Cloud-Security-in-Amazon-Web-Services

Terraform, AWS EC2, VPC, KMS, Systems Manager, Checkov

- Audited an earlier console build: SSH open to the internet, a NACL deny rule that never matched, and a dead CPU alarm.
- Rebuilt it in Terraform with no inbound SSH, IMDSv2 required, and KMS-encrypted VPC flow logs.
- Gated every push on TFLint, Checkov (121 passed, 0 failed), and 10 plan-time tests.

TrustKart | Online store github.com/VijaysinghPuwar/TrustKart

Java 21, Spring Boot, Spring Security, PostgreSQL, Redis, React

- Made checkout a single transaction that cannot duplicate orders, oversell, or deadlock under 100 concurrent shoppers.
- Secured accounts with Argon2id hashing, rotating refresh tokens, CSRF protection, and Redis rate limits.

Dual ISP Network Assessment github.com/VijaysinghPuwar/Dual-ISP-Network-Assessment-and-Subnet-Remediation

Nmap, Wireshark, PowerShell, Python

- Traced conflicting device inventories to two ISP routers on the same subnet; one gateway IP answered from two MACs.
- Moved one LAN to its own subnet and confirmed one gateway MAC, one DHCP server, and no duplicate addresses.

x86-64 Binary Hardening Lab github.com/VijaysinghPuwar/x86-64-assembly-intro-lab
NASM, ELF, gdb, readelf, GitHub Actions

- Found an executable stack and a stack alignment bug in working programs, then rebuilt them with NX, PIE, and full RELRO.
- Added a CI check that fails any binary missing those protections, tested against deliberately broken binaries.

Hardened Static Page github.com/VijaysinghPuwar/hardened-static-page
Content Security Policy, security headers, gitleaks, GitHub Actions

- Set a strict CSP and nine security headers, and wrote a CI test that injects an inline script and fails if it runs.

spoonstill | Open source video renderer github.com/VijaysinghPuwar/spoonstill
Rust, GitHub Actions, macOS and Windows

- Shipped 21 releases; CI runs 776 tests and cargo audit, and the installers verify SHA-256 before installing.

EDUCATION AND CERTIFICATIONS

M.S. Cybersecurity | Pace University, Seidenberg School of CSIS | GPA 3.92 Jan 2025 – Dec 2026 (expected)
Relevant coursework: Network Security and Defense, Ethical Hacking and Penetration Testing, Cyber Intelligence Analysis

B.E. Mechanical Engineering | G H Patel College of Engineering and Technology Aug 2023
Certifications: CompTIA Security+, CompTIA CySA+, Cisco CCNA, Google AI Essentials

SKILLS

Security: Splunk, IDS/IPS, firewalls, Burp Suite, OWASP ZAP, Nessus, SonarQube, Checkov, gitleaks, Wireshark, Nmap
Identity: Active Directory, Group Policy, DUO MFA, Zscaler ZPA
Cloud and Systems: AWS, Azure, Google Cloud Run, Cloudflare, Terraform, Ansible, Docker, Linux, Windows Server
Networking: TCP/IP, DNS, DHCP, VLANs, OSPF, BGP, VPN, Cisco Catalyst, Cisco IOS
Languages: Python, Java, TypeScript, SQL, PowerShell, Bash, C#, Rust, R
Databases: PostgreSQL, MongoDB, Redis